

Week 4 – ReactJS

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Exercise 1 (Line 31): Creating Your First React Application

Objective

Create your first React application using **Vite** (recommended) or **Create React App**, and display a simple welcome message.

Step 1: Create the Project

Open Terminal or Command Prompt:

```
npm create vite@latest my-first-react-app
```

Choose:

Framework : React

Variant : JavaScript

Move into the project:

```
cd my-first-react-app
```

Install dependencies:

```
npm install
```

Start the development server:

```
npm run dev
```

You will see an address like:

```
http://localhost:5173
```

Open it in your browser.

Project Structure

```
my-first-react-app
|
|— node_modules
|— public
|— src
|   |— App.jsx
|   |— main.jsx
|   |— App.css
```

```
|   └─ index.css
|
|   └─ package.json
└─ vite.config.js
```

Step 2: Edit App.jsx

Replace the default code with:

```
function App() {
  return (
    <div>
      <h1>Welcome to React!</h1>
      <h2>My First React Application</h2>
      <p>Hello, Java FSE Learner!</p>
    </div>
  );
}

export default App;
```

Step 3: main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

Output

When you open:

<http://localhost:5173>

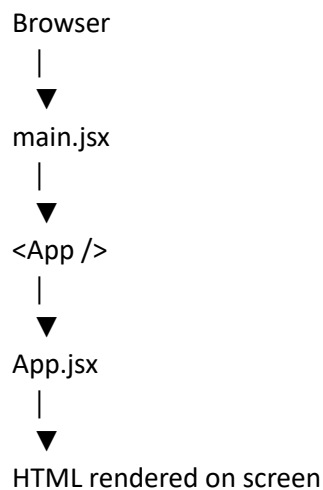
You should see:

Welcome to React!

My First React Application

Hello, Java FSE Learner!

How React Works



Important Concepts

What is React?

React is a **JavaScript library** used to build interactive user interfaces using reusable components.

What is JSX?

JSX (**JavaScript XML**) allows you to write HTML-like syntax inside JavaScript.

Example:

```
const element = <h1>Hello React!</h1>;
```

What is a Component?

A component is a reusable piece of UI.

Example:

```
function App() {  
  return <h1>Hello</h1>;  
}
```

Exercise 2 (Line 32): React Components and Props

This is one of the most important React concepts and is frequently asked in interviews.

Objective

Learn to:

- Create reusable React Components

- Pass data using **Props**
 - Render multiple components
-

Project Structure

ReactComponentsDemo

```
|
| └─ src
|   └─ components
|       └─ Student.jsx
|       └─ App.jsx
|       └─ main.jsx
|       └─ index.css
|
└─ package.json
└─ vite.config.js
```

Step 1: Create Student Component

Create a folder:

src/components

Create **Student.jsx**

```
function Student(props) {

  return (
    <div style={{
      border: "2px solid black",
      padding: "15px",
      margin: "10px"
    }}>

      <h2>Student Details</h2>

      <p><b>Name :</b> {props.name}</p>

      <p><b>Age :</b> {props.age}</p>

      <p><b>Course :</b> {props.course}</p>

    </div>
  );
}
```

```
export default Student;
```

Step 2: App.jsx

```
import Student from "../components/Student";

function App() {

  return (

    <div>

      <h1>React Components and Props Demo</h1>

      <Student
        name="Harshpreet Singh"
        age="23"
        course="Java FSE"
      />

      <Student
        name="Rahul"
        age="22"
        course="ReactJS"
      />

      <Student
        name="Aman"
        age="24"
        course="Spring Boot"
      />

    </div>

  );

}

export default App;
```

Step 3: main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
```

```
import App from "./App";

ReactDOM.createRoot(document.getElementById("root")).render(

  <React.StrictMode>
    <App />
  </React.StrictMode>

);
```

Output

React Components and Props Demo

Student Details

Name : Harshpreet Singh
Age : 23
Course : Java FSE

Student Details

Name : Rahul
Age : 22
Course : ReactJS

Student Details

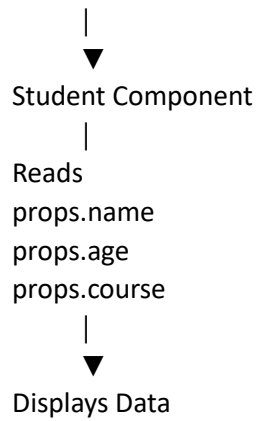
Name : Aman
Age : 24
Course : Spring Boot

Each **Student** component displays different data while using the same component code.

How Props Work

App.jsx

```
|
|
| — name="Harshpreet"
| — age="23"
| — course="Java FSE"
```



What are Props?

Props (Properties) are used to pass data from a **Parent Component** to a **Child Component**.

Example:

```
<Student
  name="Rahul"
  age="22"
  course="ReactJS"
/>
```

Inside the child component:

```
props.name
props.age
props.course
```

Parent and Child Components

Parent Child

App.jsx Student.jsx

The parent passes data, and the child receives it through props.

Why Use Components?

- Reusable code
- Better organization
- Easier maintenance
- Faster development

Instead of writing the student details three times, we create one component and reuse it with different props.

Alternative: Destructuring Props

Instead of:

```
function Student(props) {  
  
  return (  
    <h2>{props.name}</h2>  
  );  
  
}
```

You can write:

```
function Student({ name, age, course }) {  
  
  return (  
  
    <div>  
  
      <h2>{name}</h2>  
  
      <p>{age}</p>  
  
      <p>{course}</p>  
  
    </div>  
  
  );  
  
}
```

This is cleaner and commonly used in React applications.

Interview Questions

1. What is a React Component?

A React component is a reusable piece of UI that returns JSX and can be used multiple times in an application.

2. What are Props?

Props (Properties) are read-only values passed from a parent component to a child component.

3. Can a child component modify props?

No. Props are **immutable (read-only)**. If data needs to change, it should typically be managed by the parent using state.

4. Difference between Props and State

Props	State
Passed from parent	Managed inside the component
Read-only	Can be updated
Used for communication	Used for dynamic data

5. Why are components important in React?

Components make the application modular, reusable, easier to test, and easier to maintain.

Real-World Example

Imagine an e-commerce website:

Instead of creating separate HTML for every product, you create one Product component:

```
<Product name="Laptop" price={60000} />
```

```
<Product name="Mobile" price={25000} />
```

```
<Product name="Headphones" price={3000} />
```

The same component is reused with different data, just like the Student component in this exercise.